Day 4: Network Cable Types and Tools Theory (30 min): Types of cables (Ethernet, fiber optics) and tools (RJ45 crimper). Practical (30 min): Create a straight-through Ethernet cable.

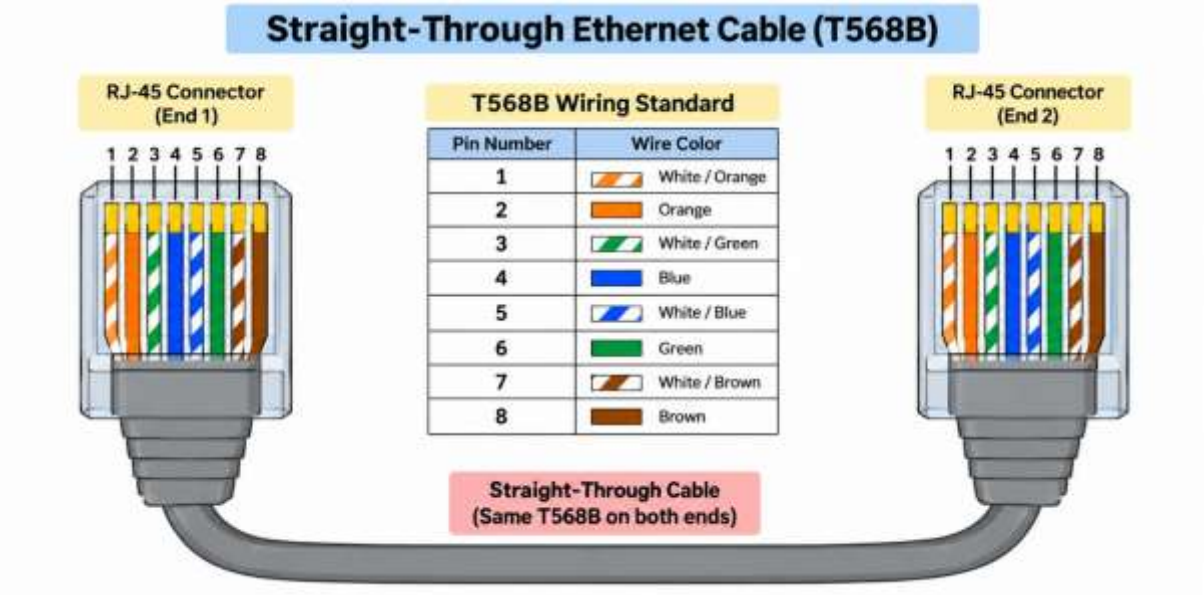
1.List 3 key differences between Ethernet cables and fiber optic cables, and give one real-world example where each type would be used (for example, home WiFi router, data center, college campus backbone).

ANS:

Ethernet Cable	**Fiber Optic Cable**
1. It uses electrical signals.	1. It uses light signals.
2. It is good for short distances.	2. It is good for long distances.
3. It is usually cheaper.	3. It is usually more expensive.

2. Draw and label a diagram showing the internal wiring of a straight-through Ethernet cable (use T568B wiring standard), indicating the color of each wire and its pin number.

ANS:



3. Watch a YouTube video demonstrating how to use an RJ45 crimper, then write a step-by-step guide (in your own words) on how to attach an RJ45 connector to a CAT6 cable.

ANS:

1. Cut the cable to the needed length.
2. Remove the cover of the cable.
3. Arrange the 8 wires in the correct color order.
4. Straighten the wires with your fingers.
5. Cut the wire ends evenly.
6. Put the wires into the RJ45 connector.
7. Use the crimper to press the connector tightly.
8. Check the wires to make sure they are in the correct order.
9. Test the cable using a cable tester.



**4.Imagine you are setting up a gaming café with 10 PCs for LAN gaming. Decide which cable type (Ethernet or fiber optics) and which tool (RJ45 crimper or fiber cleaver) you would use, and explain your choices in 3-4 sentences.

Hint: Consider speed, and budget for your answer.**

ANS:

1. I will use Ethernet cables to connect 10 PCs.
2. Ethernet cables are fast and reliable.
3. They are cheaper than fiber optic cables.
4. They are good for short distances.
5. I will use an RJ45 crimper to attach connectors.
6. This setup is easy to use and saves money.

